MILITARY SPECIFICATION

CENTRIFUGE, LABORATORY, Hand, Field

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 This specification covers a lightweight, hand-cranked laboratory centrifuge with bench-type clamp that is furnished with a 2-place head to accommodate 15 milliliter (ml.) capacity tubes. The centrifuge has a gear ratio of 16:1.

2. APPLICABLE DOCUMENTS

2.1 Specifications and standards. The following documents, of the issue in effect on date of invitation for bids or request for proposals, form a part of this specification to the extent specified herein:

SPECIFICATIONS

Federal

PPP-B-585	Boxes, Wood, Wirebound.
PPP-B-601	Boxes, Wood, Cleated-Plywood.
PPP-B-621	Boxes, Wood, Nailed and Lock-Corner.
PPP-B-636	Box, Fiberboard.

Military

MIL-M-7298	Manual, Technical: Commercial Equipment.
MIL-L-10547	Liners, Case and Sheet, Overwrap; Water-Vaporproof or Waterproof, Flexible.

FSC 6640

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STANDARDS

Military

MIL-STD-105	Sampling Procedures and Tables for Inspection by Attributes.
MIL-STD-129	Marking for Shipment and Storage.

(Copies of specifications and standards required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

3. REQUIREMENTS

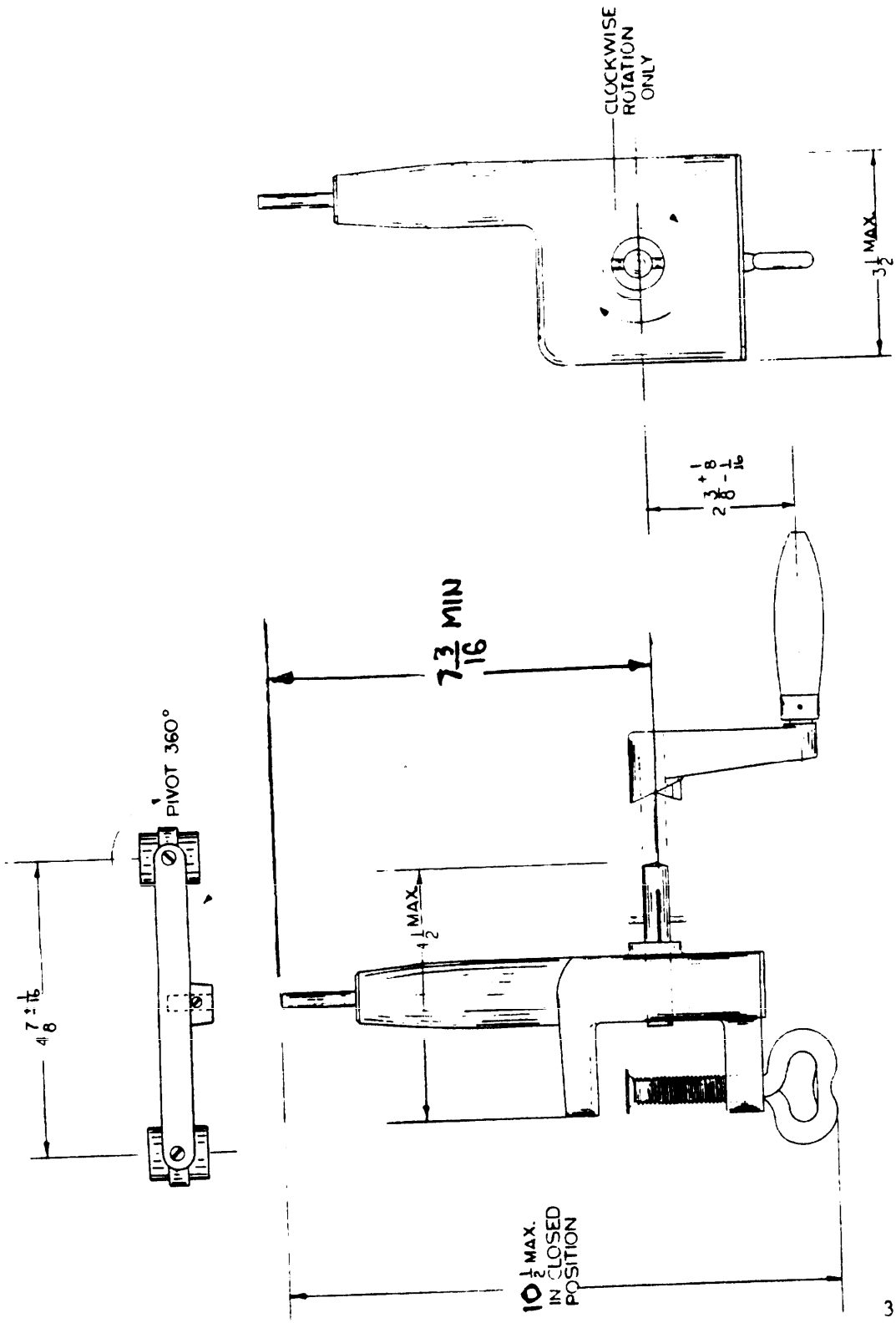
3.1 Materials. The materials used shall be suitable for intended use and shall be as hereinafter indicated.

3.2 Design and construction. The design of the centrifuge shall be in accordance with Figure 1, and shall be considered as minimum requirements. Minor variations in design shall be accepted provided they do not conflict with dimensional and configuration limitations included in the figure and specification. The centrifuge shall consist essentially of:

a. Housing with a bench clamp
b. Output shaft
c. Driving mechanism
d. Crankshaft
e. Hand crank
f. Tube head
g. Tube shields with cushions

3.2.1 Housing. The housing shall be fabricated from cast iron, except that a metal plate may be attached to the bottom of the housing. The housing shall enclose the gears and bearings of the driving mechanism (see 3.2.3), and shall have suitable cutouts for the projection of the output shaft (see 3.2.2) and crankshaft (see 3.2.4). A bench-type clamp shall be supplied as an integral part of the housing (see Figure 1); clamp shall be provided with a non-detachable thumbscrew for clamping the centrifuge unit to a working surface, plank, or tree limb having thickness up to or including 1-7/8 inches. The thumbscrew shall include a button at the contact end to facilitate clamping on wood surfaces. The thumbscrew head shall be large enough to permit tightening and loosening without the use of tools. Thumbscrew shall also be suitably plated for protection against corrosion.

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CENTRIFUGE, LABORATORY, HAND, FIELD

FIGURE 1

ALL DIMENSIONS ARE IN INCHES

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3.2.2 Output shaft. The output shaft shall be vertically mounted in the centrifuge and shall be fabricated of steel. The bottom of the output shaft within the housing shall be attached to or include a machined worm gear. The top of the output shaft shall have sufficient length above the housing enclosure to accommodate the tube head (see 3.2.6) and permit the tube shields to clear the housing when the vertically oriented output shaft is in a standstill position.

3.2.3 Driving mechanism. The driving mechanism for the rotation of the output shaft shall be enclosed within the housing. The driving mechanism shall be gear-driven and include a machined worm gear (see 3.2.2). The rotation of the gear-driven mechanism shall be controlled by the rotation of the crank shaft (see 3.2.4). When tested as specified in 4.4.1, the driving mechanism shall provide a ratio of 16 output revolutions of the output shaft for each turn of the crank shaft. The gears shall be machined and permanently lubricated. The gears, bearings, output and crank shafts shall be carefully fitted within the housing to provide a smooth and easy rotation of components (see 4.4.2).

3.2.4 Crank shaft. The crank shaft shall be horizontally mounted in the centrifuge and shall be fabricated of steel. The crank shaft shall include a pressfit pin (see Figure 1) or other suitable means to assure that the handcrank (see 3.2.5) does not slip when it is attached to and controlling the rotation of the crank shaft.

3.2.5 Hand crank. The centrifuge shall be supplied with a detachable hand crank which engages the crank shaft in such a manner as to impart only clockwise (right hand) rotation as indicated in Figure 1. The hand crank shall consist of a metal lever arm with a handle grip fabricated of hardwood or plastic having a smooth finish. The handle grip shall rotate freely. The contact end of the lever arm shall be designed so that it:

a. Engages the crank shaft without slipping while driving the rotating crank shaft.

b. Disengages the rotating crank shaft when the manual rotation is stopped.

3.2.6 Tube head. A demountable two-place tube-head shall be supplied for attachment to the output shaft. The frame of the tube head shall be fabricated of aluminum and shall support a freely rotating trunnion ring at each end. The head shall be attached to the output shaft by means of a sleeve that is integrally mounted to the bottom side of the frame, and then locked to the output shaft by means of a set screw mounted in the sleeve. The trunnion rings shall be fabricated of corrosion-resistant steel or brass suitably plated to protect against corrosion. The trunnion rings shall accommodate tube shields specified in 3.2.7.

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.2.7 Tube shields. The tube shields shall be fabricated of corrosion-resistant steel or aluminum, and shall be suitable for accommodating centrifuge tubes having a 118 millimeter (mm.) nominal length and an outside diameter of 16.8 ± 0.5 mm. (see 6.3). The tube shields shall be supplied with rubber cushions.

3.3 Operating characteristics. When tested as specified in 4.4.2, the centrifuge components shall operate in accordance with the following:

a. The hand crank and output shaft rotate easily without any obstructions or jamming.

b. The rotating output shaft is free of wobble.

c. The tube head remains properly secured to the output shaft.

d. The tubes clear the tube head when rotating freely.

e. The shields clear the operator's hand at all speeds of rotation.

f. The handle grip remains properly secured to the hand crank.

3.4 Weight. The combined weight of the centrifuge unit, hand crank, tube head and tube shields shall not exceed 4 pounds.

3.5 Finish. The exposed surfaces of the housing tube head frame and lever arm shall be finished in a suitable non-reflecting dark color.

3.6 Workmanship. Workmanship shall be first class throughout. The centrifuge shall be free from defects which detract from its appearance or may impair its serviceability.

3.7 Service data. The contractor shall furnish, with each unit, two copies each of a technical manual which contains complete instructions for installation, operation, and maintenance and lists of component parts. The manual shall comply with the requirements of MIL-M-7298. As soon as practicable after award of the contract or purchase order, the contractor shall furnish to the contracting officer, via the cognizant Government inspector, two preliminary copies of his proposed technical manual, for review and comment by the Directorate of Medical Materiel, Defense Personnel Support Center. The preliminary manuals shall be submitted sufficiently early to permit adequate review by the Directorate of Medical Materiel, Defense Personnel Support Center (based on a maximum period of 30 days for such review) and to allow incorporation by the contractor of any

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required revisions and corrections in the final manual without delaying delivery under the contract or purchase order. In addition, the contractor shall furnish 15 copies of the approved technical manual to the contracting officer for distribution. Should the supplier have furnished acceptable manuals for the identical item within two years preceding the date of contract and propose to furnish manuals identical with those previously accepted, the requirements for review of drafts and submission of 15 additional copies may be waived upon request to the contracting officer. Waiver of submission of 15 additional copies shall be the option of the Government.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or purchase order, the supplier may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.1.1 Records. Records of examinations and tests performed by or for the contractor shall be maintained by the contractor and made available to the Government, upon the Government's request, at any time, or from time to time, during the performance of the contract and for a period of 3 years after delivery of the supplies to which such records relate.

4.1.2 Inspection. Inspection, as used in this specification, is defined as both examination (such as visual, or auditory, investigation without the use of special laboratory appliances or procedures) and testing (determination by technical means of physical and chemical properties) of the item.

4.2 Sampling.

4.2.1 For examination. Sampling for examination shall be conducted in accordance with MIL-STD-105 and Table I.

Table I. For examination

	Inspection level	AQL (percent defective)
For visual examination		
Major defects	II	1.0
Minor defects	II	2.5
For dimensional examination	S-3	2.5
For weight examination	S-3	2.5

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4.2.2 For tests. Sampling for examination shall be conducted in accordance with MIL-STD-105 and Table II. The acceptance number shall be zero.

Table II. For tests

Characteristic	Requirement	Test procedure	Inspection level
Gear ratio	3.2.3	4.4.1	S-2
Operation	3.3	4.4.2	S-2

4.3 Examination. The centrifuge shall be examined to determine compliance with all requirements contained in this specification.

4.3.1 Classification of defects. Examination shall be conducted in accordance with Table III. Examination shall not be restricted to the classified possible defects.

Table III. Classification of defects

Categories	Defects
Major	
101	Centrifuge components not free of sharp edges or burrs.
102	Crank handle not capable of being properly attached to or detached from crank shaft.
103	Thumb screw not capable of being adjusted to desired positions.
104	Component parts not free of corrosion.
105	Output and crank shafts not properly aligned in housing.
106	Trunnion rings not properly aligned in tube head.
107	Tube head not capable of being properly attached to or detached from output shaft.
108	Handle surface not smooth.
109	Tube shields not free of distortion, or not capable of being attached to or detached from trunnion rings.
Minor	
201	Finished surfaces not free of blistering, peeling or flaking.
202	Component surfaces (other than handle) not smooth.
203	Identification marking not present, complete or legible.

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4.3.2 Dimensional and weight examination. The centrifuge shall be examined for defects in dimensions and weight limitation. Any dimension or weight not within the limits specified herein shall be classified as a defect.

4.3.3 Examination for preparation for delivery. An examination shall be made to determine whether the packaging, packing and marking comply with the requirements of Section 5. Defects shall be scored as specified in Table IV. Sampling shall be in accordance with MIL-STD-105. The sample unit shall be one container fully prepared for delivery. The lot shall be the number of containers offered for inspection at one time. The inspection level shall be S-2, and the acceptable quality level shall be 4.0 defects per hundred units.

Table IV. Examination for preparation for delivery

Examination	Defect
Containers	Not as specified.
Quantity	Quantity in interior packages and shipping containers not as specified.
Materials	Any component missing or damaged.
Workmanship	Inadequate application of components, such as incomplete closure of container flaps, tape improperly applied, or containers distorted.
Markings	Omitted, incorrect, illegible, improper size, location, sequence or method of application.

4.4 Tests. Tests shall be conducted to determine compliance with specification requirements.

4.4.1 Gear ratio. A hand crank shall be attached to the crank shaft. The number of rotations of the output shaft for each complete revolution of the hand crank shall be determined and recorded.

4.4.2 Operating characteristic. A tube head with a tube shield inserted in each trunnion ring shall be securely attached to the output shaft of the centrifuge. The tube shield shall be supplied with centrifuge tubes conforming to 3.2.7. The centrifuge shall be clamped to a suitable surface. A hand crank shall be attached to the crank shaft. The handle grip shall be grasped in such a manner as to keep the operator's hand clear of the tube shields. The hand crank shall be turned with sufficient speed so that the tube shields are in the horizontal position, and then checked for any evidence of improper operation.

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5. PREPARATION FOR DELIVERY

5.1 Preservation and packaging. Preservation and packaging shall be Level A or C as specified (see 6.1).

5.1.1 Level A.

5.1.1.1 Preservation. All exposed bare metal surfaces other than those made of corrosion resistant steel shall be coated with a suitable preservative for protection against corrosion. All preserved items shall be wrapped in greaseproof barrier material and the wrap secured.

5.1.1.2 Packaging. Each centrifuge preserved as specified in 5.1.1.1 shall be packaged in a snug-fitting box constructed in accordance with PPP-B-636, type CF, class domestic. The box shall include sufficient cushioning material to insure against shifting of or damage to the contents. Closure of the box shall be secured to prevent accidental opening.

5.1.3 Level C. Units shall be packaged in standard commercial containers of the size and kind commonly used, which will afford the degree of protection required for shipment and use of the product for its intended purpose.

5.2 Packing. Packing shall be Level A, B or C as specified (see 6.1).

5.2.1 Level A. Ten unit packages shall be packed in an exterior container designed for a type 2 load and constructed in accordance with PPP-B-585, class 3, style 3; PPP-B-601, overseas type; PPP-B-621, class 2; or PPP-B-636, class weather-resistant. Closure and strapping shall be as specified in the appendix of the applicable box specification.

5.2.1.1 Case liner. Each Level A wood box shall be lined with a waterproof case liner conforming to MIL-L-10547. Closure and sealing shall conform to applicable paragraphs of appendix thereto. Case liner shall not be required for fiberboard boxes.

NOTE: Strapping shall not be required for shipments forwarded to a receiving activity within the continental limits of the United States for storage and redistribution.

5.2.2 Level B. Ten unit packages shall be packed in an exterior container designed for a type 2 load and constructed in accordance with PPP-B-585, class 1, style 3; PPP-B-601, domestic type; PPP-B-621, class 1; or PPP-B-636, class domestic. Closure of wood boxes shall be as specified in the appendix of the applicable box specification. Closure of fiberboard boxes shall conform to Method II of PPP-B-636.

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5.2.3 Level C. The subject commodity shall be packed in substantial commercial containers of the type, size, and kind commonly used for the purpose, so constructed as to insure acceptance and safe delivery by common or other carriers, at the lowest rate, to point of delivery called for in the contract or purchase order.

5.2.4 Packing variation permitted. If the required number of units in the entire shipment is less than the number of units specified to be overpacked in an exterior container, such units may be packed in an exterior container of suitable size and design, acceptable to a common carrier, which shall insure safe delivery to destination.

5.3 Marking.

5.3.1 Unit packages. Each unit package shall be marked in accordance with MIL-STD-129. When labels are utilized, waterproofing shall be required only when applicable carton is fabricated of water-resistant material. Contract or purchase order number and name of contractor shall be shown.

5.3.2 Exterior container. Exterior container shall be marked in accordance with MIL-STD-129.

6. NOTES

6.1 Ordering data. Procurement documents should specify the following:

a. Title, number and date of this specification.
b. National Stock Number (NSN).
c. Applicable levels of packaging and packing (see 5.1 and 5.2).

6.2 Intended use. The centrifuge is intended to fit into an intermediate container case which is part of a medical laboratory equipment set. The container case and equipment set are listed in the Federal Supply Catalog as indicated:

National Stock No.	Item Identification
6545-00-926-9037	CASE, CENTRIFUGE-MICROSCOPE-REAGENTS, Empty
6545-00-926-1483	MEDICAL EQUIPMENT SET, MEDICAL LABORATORY, Combat, Individual

6.3 The tube shields are intended to be suitable for use with tubes listed in the Federal Supply Catalog as indicated:

National Stock No.	Item Identification
6640-00-926-6986	TUBE, CENTRIFUGE, Polyethylene, Conical Bottom, Graduated, 15 ml, 12s

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6.4 The centrifuge covered by this specification is listed in the Federal Supply Catalog as indicated:

National Stock No.	Item identification
6640-00-926-6913	CENTRIFUGE, LABORATORY, Hand, Field

Custodians:
Army - MD
Navy - MS
Air Force - 03

Preparing activity:
Defense Supply Agency - DM

Project No. 6640-0912
CS

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TANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

OMB Approval
No. 22-R255

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DOCUMENT IDENTIFIER AND TITLE
MIL-C-37099 CENTRIFUGE, LABORATORY, Hand, Field

NAME OF ORGANIZATION AND ADDRESS

CONTRACT NUMBER

MATERIAL PROCURED UNDER A
☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE DOCUMENT CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?
A. GIVE PARAGRAPH NUMBER AND WORDING.

B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

2. COMMENTS ON ANY DOCUMENT REQUIREMENT CONSIDERED TOO RIGID

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☐ YES ☐ NO *(If "Yes", in what way?)*

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